Tech Bite #6: Components -- Resistors, Capacitors, Inductors, and Semiconductors

The basic parts every radio is built from -- what each one does. You do not design circuits; you recognize the pieces.

Groups T6A + T6B · 23 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

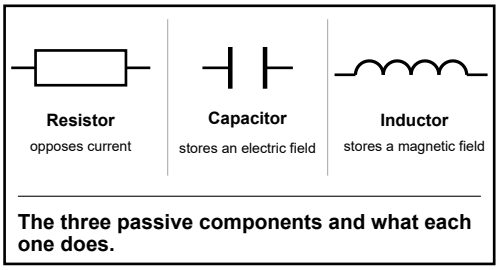
The Big Picture

Now we move from electricity itself to the **parts** that shape it. This Bite covers the basic components – the resistors, capacitors, inductors, and semiconductors that every radio is built from. You do not need to design circuits; you just need to know what each part does.

The Three Passive Components

Three components turn up everywhere:

- A **resistor** opposes the flow of current. (T6A01) A **potentiometer** is an adjustable resistor – it **controls resistance**, and it is what a volume knob usually is. (T6A02, T6A03)
- A **capacitor** stores energy in an **electric field**. (T6A04) It is two conductive surfaces separated by an insulator. (T6A05)
- An **inductor** stores energy in a **magnetic field**. (T6A06) It is typically just a coil of wire. (T6A07)



The three passive components and what each one does.

Switches, Fuses, and Batteries

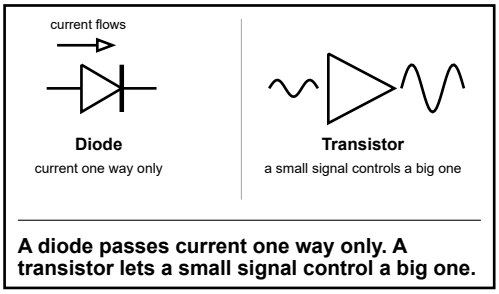
A **switch** makes or breaks a circuit. An **SPDT** (single-pole double-throw) switch routes one circuit to **either of two others**. (T6A08, T6A09) Batteries come in two kinds: **rechargeable** – nickel-metal hydride, lithium-ion, and lead-acid are all rechargeable (T6A10) – and **non-rechargeable**, like the common carbon-zinc cell. (T6A11)

Semiconductors: Diodes and Transistors

Semiconductors are the **active** parts.

A **diode** lets current flow in **only one direction** (T6B02) – like a one-way valve. Its two electrodes are the **anode and cathode** (T6B09); the cathode end is marked with a **stripe**. (T6B06) A diode has a small **forward voltage drop**, lower in some types than others. (T6B01) An **LED** emits light when **forward current** flows through it. (T6B07)

A **transistor** has **three regions** of semiconductor material (T6B04) and works as an electronic **switch** (T6B03) or an **amplifier**. **Gain** is the ratio of output to input. (T6B10, T6B11)



A diode passes current one way only. A transistor lets a small signal control a big one.

The two common transistor types: the **bipolar junction transistor**, with electrodes called **emitter, base, and collector** (T6B12), and the **FET** – **Field Effect Transistor** (T6B08) – with a **gate, drain, and source**. (T6B05)

Mental Cheat Sheet

- **Resistor** opposes current; a **potentiometer** is an adjustable resistor.
- **Capacitor** stores energy in an **electric** field.
- **Inductor** (a coil) stores energy in a **magnetic** field.
- **SPDT** switch routes one circuit to either of two others.
- Rechargeable: NiMH, lithium-ion, lead-acid. **Carbon-zinc is not.**
- **Diode**: current one way only; electrodes are **anode and cathode**; cathode = stripe.
- **Transistor**: 3 regions; works as a switch or amplifier; **gain** = output / input.
- **BJT**: emitter, base, collector. **FET**: gate, drain, source.

Common Traps

- Capacitor stores an **electric** field; inductor stores a **magnetic** field – do not swap.
- A potentiometer controls **resistance** – not capacitance or inductance.
- **Carbon-zinc** is the non-rechargeable one in the list.
- A diode's electrodes are **anode and cathode** – not "source and drain" (that is a FET).
- The **cathode** is the striped end of a diode.
- An LED lights on **forward** current, not reverse.

Takeaway: Every radio is built from a few parts. The **passives** – resistor (opposes current), capacitor (stores an electric field), inductor (stores a magnetic field) – shape the current. The **semiconductors** – the diode (one-way) and the transistor (switch or amplifier) – are the active parts that make a radio do real work. You only need to recognize each one and know its job.

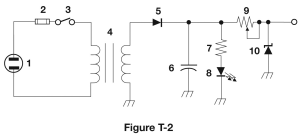


Figure T-2

Figure T-2 – used by question T6A09.

Bite 6 Question Bank (T6A + T6B)

T6A01. What electrical component opposes the flow of current in a DC circuit?
A. Inductor
B. Resistor
C. Inverter
D. Transformer

T6A02. What type of component is often used as an adjustable volume control?
A. Fixed resistor
B. Power resistor
C. Potentiometer
D. Transformer

T6A03. What electrical parameter is controlled by a potentiometer?
A. Inductance
B. Resistance
C. Capacitance
D. Field strength

T6A04. What electrical component stores energy in an electric field?
A. Resistor
B. Capacitor
C. Inductor
D. Diode

T6A05. What type of electrical component consists of conductive surfaces separated by an insulator?
A. Resistor
B. Potentiometer
C. Oscillator
D. Capacitor

T6A06. What type of electrical component stores energy in a magnetic field?
A. Resistor
B. Capacitor
C. Inductor
D. Diode

T6A07. What electrical component is typically constructed as a coil of wire?
A. Transistor
B. Capacitor
C. Diode
D. Inductor

T6A08. What is the function of an SPDT switch?
A. A single circuit is opened or closed
B. Two circuits are opened or closed
C. A single circuit is switched between one of two other circuits
D. Two circuits are each switched between one of two other circuits

T6A09. What type of switch is represented by component 3 in figure T-2?
A. Single-pole single-throw
B. Single-pole double-throw
C. Double-pole single-throw
D. Double-pole double-throw

T6A10. Which of the following battery chemistries is rechargeable?
A. Nickel-metal hydride
B. Lithium-ion
C. Lead-acid
D. All these choices are correct

T6A11. Which of the following battery chemistries is not rechargeable?
A. Nickel-cadmium
B. Carbon-zinc
C. Lead-acid
D. Lithium-ion

T6B01. Which is true about forward voltage drop in a diode?
A. It is lower in some diode types than in others
B. It is proportional to peak inverse voltage
C. It indicates that the diode is defective
D. It has no impact on the voltage delivered to the load

T6B02. What electronic component allows current to flow in only one direction?
A. Resistor
B. Fuse
C. Diode
D. Driven element

T6B03. Which of these components can be used as an electronic switch?
A. Varistor
B. Potentiometer
C. Transistor
D. Thermistor

T6B04. Which of the following components can consist of three regions of semiconductor material?
A. Alternator
B. Transistor
C. Triode
D. Pentode

T6B05. What type of transistor has a gate, drain, and source?
A. Varistor
B. Field-effect
C. Hall-effect
D. Bipolar junction

T6B06. How is the cathode lead of a semiconductor diode often marked on the package?
A. With the word “cathode”
B. With a stripe
C. With the letter C
D. With the letter K

T6B07. What causes a light-emitting diode (LED) to emit light?
A. Forward current
B. Reverse current
C. Capacitively-coupled RF signal
D. Inductively-coupled RF signal

T6B08. What does the abbreviation FET stand for?
A. Frequency Emission Transmitter
B. Fast Electron Transistor
C. Free Electron Transmitter
D. Field Effect Transistor

T6B09. What are the names for the electrodes of a diode?
A. Plus and minus
B. Source and drain
C. Anode and cathode
D. Gate and base

T6B10. Which of the following can provide power gain?
A. Transformer
B. Transistor
C. Reactor
D. Resistor

T6B11. What does the term gain mean in amplifiers?
A. The output signal voltage relative to the input signal voltage
B. The output signal current relative to the input signal current
C. The output signal power relative to the input signal power
D. All these choices are correct

T6B12. What are the names of the electrodes of a bipolar junction transistor?
A. Signal, bias, power
B. Emitter, base, collector
C. Input, output, supply
D. Pole one, pole two, output

Answer Key

T6A01: B	T6A02: C	T6A03: B	T6A04: B	T6A05: D
T6A06: C	T6A07: D	T6A08: C	T6A09: A	T6A10: D
T6A11: B	T6B01: A	T6B02: C	T6B03: C	T6B04: B
T6B05: B	T6B06: B	T6B07: A	T6B08: D	T6B09: C
T6B10: B	T6B11: D	T6B12: B		